

8086 CPU Architecture & Internal Registers

Q. Explain the internal architecture of the 8086 microprocessor. Describe the functions of the Bus Interface Unit (BIU) and the Execution Unit (EU).

[DEF] Definition to Remember

The Intel 8086 is a 16-bit microprocessor with a 20-bit address bus (1 MB memory). Its architecture is divided into two asynchronously operating units: the Bus Interface Unit (BIU) (handles memory/bus fetches) and the Execution Unit (EU) (decodes and executes instructions), enabling early instruction pipelining.

1. Bus Interface Unit (BIU)

The BIU handles all data and address transfers on the system buses.

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2. Execution Unit (EU)

The EU is responsible for decoding and executing the fetched instructions.

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3. Parallel Operation

Because the BIU and EU operate independently, the BIU continuously fills the 6-byte queue from memory while the EU executes the previously fetched instructions. This prevents the EU from sitting idle waiting for memory fetches, drastically reducing execution time.

[MUST-WRITE] Must-Write Points (for 10 marks)

1. 8086 architecture is divided into the BIU (Bus Interface Unit) and EU (Execution Unit).
2. BIU handles memory/bus operations and calculates the 20-bit physical address.
3. BIU contains Segment Registers (CS, DS, SS, ES) and a 6-byte Instruction Queue for pipelining.
4. EU decodes and executes instructions from the queue.
5. EU contains the ALU, General Purpose Registers (AX, BX, CX, DX), Pointers, and Flags.
6. The independent operation allows the BIU to prefetch instructions while the EU executes them.
7. This parallel operation minimizes CPU idle time.

[RECALL] Quick Recall

8086 Arch -> BIU + EU -> BIU (Bus transfers, Address Calc, 6-byte Queue, Seg Registers) -> EU (Execute, ALU, Gen Registers, Flags) -> Parallel operation (Pipelining)